

YOURSTORY



1125



CoCo CLI

HACKATHON

GCC Edition



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Team Size : 1

Problem Statement : Predictive Maintenance and OEEE Command Centre

Submission Guidelines:

(Things to add on the submission deck)

1. Problem Brief

- What real business problem does this solve?
- Who is the target user/persona?
- What is the current pain point and how does this improve it?
- Industry/domain context

2. Architecture Diagram

- System design showing data flow
- Which CoCo CLI skills are used and how they connect
- Data sources (structured/unstructured)
- How modular components plug together

3. Impact Statement

- Measurable outcomes (time saved, accuracy improvement, etc.)
- Scalability potential
- How this extends beyond the demo

1. Problem Brief

- What real business problem does this solve?

Unplanned machine failures can lead to production downtime, maintenance delays and operational losses. In many manufacturing environments, sensor data is available but is not easily translated into actionable maintenance decisions. MFG Sentinel uses machine sensor readings to predict failure risk in advance, identify likely failure modes and provide actionable insights so maintenance teams can intervene proactively rather than reacting after a breakdown.

The approach is implemented on Snowflake with a foundation for future ERP, maintenance and OEE integration.

- Who is the target user/persona?

Plant Managers, Maintenance Engineers, Reliability Engineers and Operations Teams.

- What is the current pain point and how does this improve it?

Maintenance teams often react to failures after disruption occurs. MFG Sentinel converts sensor readings into failure probability, diagnosis, recommended action and alerts, enabling more proactive maintenance decisions.

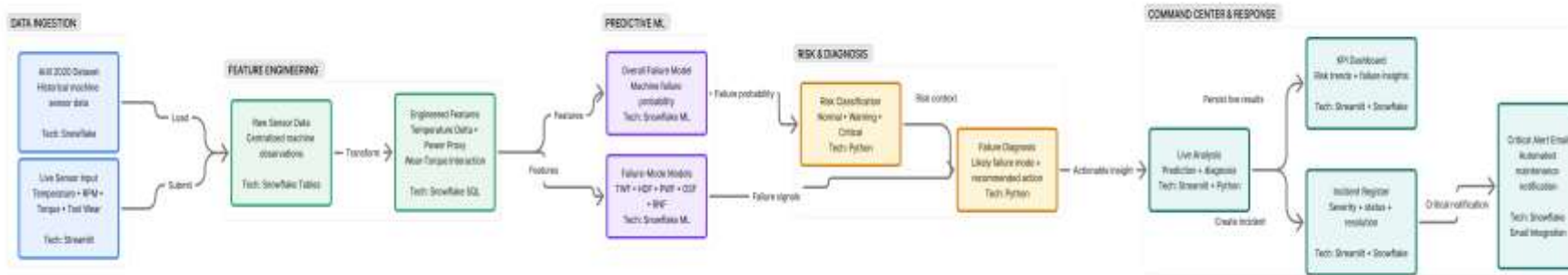
- Industry/domain context

Manufacturing • Industrial IoT • Predictive Maintenance

2. Architecture Diagram

- System Design & Data Flow

The MFG Sentinel architecture is designed as a modular, Snowflake-centric predictive-maintenance pipeline, where data ingestion, ML prediction, diagnosis, monitoring and incident management are connected through a common analytical layer.



2. Architecture Diagram

- Which CoCo CLI skills are used and how they connect

- > **Snowflake SQL / Data Engineering:**

Used for data ingestion, transformation and feature engineering.

- > **Snowflake ML:**

Used to create, train, evaluate and run the six predictive-maintenance models.

- > **Streamlit / Python:**

Used to build the interactive Command Center, live prediction workflow, KPI views and incident management.

2. Architecture Diagram

- Data sources

The current MVP uses the AI4I 2020 Predictive Maintenance Dataset, containing 10,000 machine observations and 14 attributes. The attributes fall into three categories:

1. **Machine Context:** UDI, Product ID and product Type provide basic observation and product information.
2. **Sensor Signals:** Air/Process Temperature, RPM, Torque and Tool Wear represent the machine's operating condition.
3. **Failure Labels:** The dataset contains Machine Failure and five specific failure-mode labels:-
 - **TWF:** Tool Wear Failure
 - **HDF:** Heat Dissipation Failure
 - **PWF:** Power Failure
 - **OSF:** Overstrain Failure
 - **RNF:** Random Failure

- How modular components plug together

2. Architecture Diagram

- How modular components plug together

- **DataLayer** → **ML Layer**: Snowflake tables and SQL feature engineering prepare sensor data into model-ready features.
- **ML Layer** → **Intelligence Layer**: Six Snowflake ML models generate overall failure probability and failure-mode predictions.
- **Intelligence** → **Application**: Python-based diagnostic logic converts model outputs into risk level, likely failure mode and recommended action.
- **Application Layer**: Streamlit brings these outputs together across Live Analysis, KPI Dashboard and Incident Register.
- **Action Layer**: Critical-risk results create an incident and trigger automated email notification through Snowflake.

Overall:

Ingest → Engineer → Predict → Diagnose → Visualize → Alert & Track

3. Impact Statement

- Measurable outcomes:

1. 90.7% Macro F1 and 97.9% Macro AUC for overall failure prediction.
2. Automated diagnosis, KPI monitoring and critical-risk email alerts reduce manual intervention.

- Scalability potential

Handles high-volume machine logs and sensor observations by leveraging Snowflake's scalable storage and compute architecture for large-scale ingestion, processing and ML inference.

- How this extends beyond the demo

1. Extend to real-time IoT streams, ERP/maintenance data, OEE monitoring and automated work-order workflows.
2. Include machine/tool specific maintenance records for better action predictions specific to tools/machines.

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THANK YOU